

Hyunjae Suh

Ph.D. Candidate in Software Engineering
University of California, Irvine

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RESEARCH INTERESTS

AI for Software Engineering · Large Language Models · AI-generated Code Attribution & Provenance · Code Generation Evaluation · AI-assisted Software Development · Empirical Software Engineering

RESEARCH EXPERIENCE

Ph.D. Researcher – AI & Software Engineering
University of California, Irvine

Sep. 2023 – Present
Irvine, CA

- Conduct research on applying and evaluating large language models for software engineering tasks, focusing on AI-generated code attribution, accessible code generation, and reliable AI-assisted development.
- Investigated automatic attribution of AI-generated source code using semantic code embeddings, fine-tuned language models, and feature-based classifiers, demonstrating the effectiveness of semantic program representations and achieving an 82.55 F1-score. Published at **ICSE 2025**.
- Conducted empirical studies on improving accessibility-aware code generation using prompting strategies including Zero-shot, Few-shot, Self-Criticism, and proposed *FeedAI1y*, an accessibility-aware iterative refinement framework integrating automated accessibility evaluation into LLM workflows. Published at **ACM TOSEM**.
- Designed large-scale experimental pipelines for evaluating LLM behavior across multiple models, prompting strategies, datasets, and software engineering tasks, emphasizing reproducibility and statistically rigorous empirical evaluation.

INDUSTRY RESEARCH EXPERIENCE

Applied Scientist Intern
Amazon

Jun. 2026 – Sep. 2026
Seattle, WA

- Investigating machine learning approaches for large-scale workload forecasting in AWS infrastructure systems, focusing on improving transaction-per-second (TPS) prediction accuracy.
- Developed data-driven business driver alignment methods to identify service-specific predictors of TPS, evaluating causal and statistical relationships among operational signals.
- Explored forecasting architectures and feature modeling strategies to improve prediction robustness beyond existing business-driver-based approaches.

Applied Scientist Intern
Amazon

Jun. 2025 – Sep. 2025
Austin, TX

- Investigated deployment risk assessment for large-scale software systems by analyzing code changes in Prime Video production repositories, developing machine learning approaches to identify potentially risky deployments.
- Designed a diff-aware feature framework that extracts code-level, change-level, and qualitative signals from software modifications to improve deployment risk prediction.
- Investigated the capability of LLMs for software change risk assessment by integrating diff-aware features into LLM-based classification pipelines and evaluating their effectiveness across production and open-source datasets.
- Demonstrated that carefully curated structural code complexity features provide stronger risk signals than traditional change-volume metrics, achieving an average F1-score of 0.81 across evaluated datasets.
- Drove the research project from idea development through experimentation and publication, resulting in acceptance at the **ASE 2026 Industry Showcase**.

Graduate Research Assistant
eBay

Aug. 2023 – Dec. 2023
Remote

- Investigated automated commit message generation using instruction-tuned open-source large language models for software engineering workflows.
- Evaluated prompt engineering and supervised fine-tuning strategies for repository-specific commit message generation across more than 50 software repositories.
- Improved commit message quality and stylistic consistency through instruction tuning and empirical evaluation of multiple open-source LLMs.

PUBLICATIONS

* indicates equal contribution.

An Empirical Study on Evaluating Accessible Code Generation Capabilities of LLMs

Hyunjae Suh, Mahan Tafreshipour, Sam Malek, Iftekhar Ahmed

ACM Transactions on Software Engineering and Methodology (TOSEM), 2026. [\[DOI\]](#)

Deployment Risk Assessment Using Diff-Aware Features: A Case Study at Prime Video

Mayur Kurup*, **Hyunjae Suh***, Swathi Vaidyanathan, Pranesh Vyas, Srinidhi Madabhushi, Yegor Silyutin

IEEE/ACM International Conference on Automated Software Engineering (ASE), Industry Showcase, 2026. [\[arXiv\]](#)

An Empirical Study on Automatically Detecting AI-Generated Source Code: How Far Are We?

Hyunjae Suh, Mahan Tafreshipour, Jiawei Li, Adithya Bhattiprolu, Iftekhar Ahmed

IEEE/ACM International Conference on Software Engineering (ICSE), 2025. [\[DOI\]](#)

Does the Order of Fine-tuning Matter and Why?

Qihong Chen, Jiawei Li, **Hyunjae Suh**, Lianghao Jiang, Zheng Zhou, Jingze Chen, Jiri Gesi, Iftekhar Ahmed

arXiv preprint, 2024.

EDUCATION

University of California, Irvine

Ph.D. in Software Engineering

Sep. 2023 – Present

Irvine, CA

Kookmin University

B.S. in Computer Science

Mar. 2017 – Aug. 2023

Seoul, South Korea

PROFESSIONAL SERVICE

Reviewer

ACM Transactions on Software Engineering and Methodology (TOSEM)

Empirical Software Engineering (EMSE)

IEEE Transactions on Reliability

TEACHING EXPERIENCE

Teaching Assistant

University of California, Irvine

Sep. 2023 – Present

Irvine, CA

- Supported instruction for undergraduate and graduate courses including Software Testing, Analysis, and Quality Assurance; Information Retrieval; Programming with Software Libraries; and How Computers Work.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C++ , SQL

Machine Learning: PyTorch, Hugging Face Transformers, scikit-learn, XGBoost, Keras

Large Language Models: LLM Fine-tuning, Instruction Tuning, Prompt Engineering, LLM Evaluation, Code Representation Learning, RAG

Software Engineering Research: Program Analysis, Code Embeddings, Static Analysis, Software Metrics, Empirical Software Engineering

Cloud & Infrastructure: AWS (SageMaker, Bedrock, S3), Docker, Git, LaTeX

Software Engineering Tools: Soot, Understand, Checkstyle, ESLint, Detekt